EST 113: Learning Theories for ICT Teachers

Lesson Two: Behaviorism

Behaviorism is a psychological theory and approach that emphasizes the study of observable behaviors rather than internal mental states. It is rooted in the belief that all behaviors are acquired through interaction with the environment, behaviorism posits that learning occurs as a result of conditioning—a process where associations are made between stimuli and responses. Behaviorism disregards introspection and focuses on objective, measurable outcomes, making it a scientific approach to understanding human and animal behavior. The theory was developed in the early 20th century by psychologists such as John B. Watson, who is considered the founder of behaviorism, and later expanded by B.F. Skinner. Behaviorists argue that behavior can be studied systematically and modified through reinforcement and punishment.

Behaviorism is best summed up by the following quote from Watson, who is often considered the father of behaviorism: " Give me a dozen healthy infants, well-formed, and my own specified world to bring them up in and I'll guarantee to take any one at random and train him to become any type of specialist I might select—doctor, lawyer, artist, merchant-chief and, yes, even beggar-man and thief, regardless of his talents, abilities and race of his ancestors"

Types of Behaviorism

Methodological Behaviorism: This approach emphasizes the scientific study of behavior and rejects the study of internal mental processes, focusing instead on observable behaviors and their relationships with environmental stimuli. Methodological behaviorism assumes that psychology should be an objective science, studying only what can be directly observed and measured. John B. Watson is a key figure in this approach.

Radical Behaviorism: This approach, pioneered by B.F. Skinner, views behavior as a function of environmental factors and consequences, emphasizing the role of reinforcement and punishment in shaping behavior. Radical behaviorism suggests that internal mental states are not necessary to explain behavior, focusing instead on the relationship between stimuli, responses, and consequences.

Behaviorist and Their Experiments

John B. Watson's Experiments on Behaviorism

John B. Watson, often regarded as the "father of behaviorism," revolutionized psychology by focusing on observable behaviors and dismissing the study of internal mental processes. Watson believed that all behaviors, even emotional reactions, are learned through interactions with the environment. His most famous experiment, the "Little Albert Experiment", demonstrated how classical conditioning could be applied to human emotions. Below is an extensive discussion of his key experiment and its implications.

The Little Albert Experiment (1920)

Watson aimed to show that emotional responses, such as fear, could be conditioned in humans. He sought to prove that emotions were not innate but learned through associations with external stimuli, similar to the principles of classical conditioning discovered by Pavlov.

Experiment Details

- A healthy, 9-month-old boy named "Albert B." (later referred to as "Little Albert"). Albert was chosen for his calm temperament and lack of pre-existing fears of animals or objects.
- Watson tested Albert's baseline emotional reactions to various stimuli, including a white rat, a rabbit, a monkey, masks, and burning paper.
- Albert showed no fear of these objects, indicating that his fear responses would need to be conditioned.
- Watson paired a **neutral stimulus** (a white rat) with an **unconditioned stimulus** (a loud, startling noise made by striking a steel bar with a hammer).
- When Albert reached out to touch the white rat, Watson would make the loud noise, causing Albert to cry and show signs of fear (an **unconditioned response**).
- This pairing was repeated several times over several sessions.
- After repeated pairings, Albert began to show fear (**conditioned response**) at the mere sight of the white rat (**conditioned stimulus**), even in the absence of the loud noise.
- This demonstrated that Albert had learned to associate the previously neutral rat with the fear-inducing noise.
- Watson observed that Albert's fear extended to other similar objects, such as a rabbit, a fur coat, and a Santa Claus mask with a white beard. This phenomenon, known as **stimulus generalization**, showed that conditioned responses could transfer to similar stimuli (see figure 1).

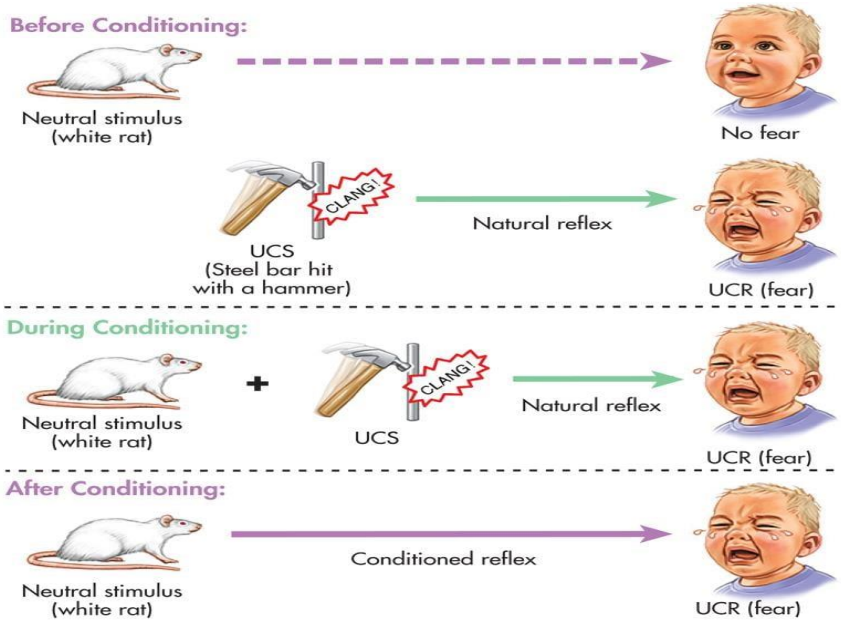


Figure 1: The little Albert Experiment

Findings of the Experiment

1. **Learned Fear Responses:**

Emotional reactions such as fear can be conditioned in humans through associations between stimuli.

2. **Stimulus Generalization:**

   A conditioned response can generalize to stimuli that are similar to the original conditioned stimulus.

3. **Implications for Behaviorism:**

   The experiment provided empirical support for Watson's belief that behavior (including emotions) could be scientifically studied and manipulated through conditioning. This laid the foundation for behaviorism as a dominant approach in psychology.

Criticisms and Ethical Concerns

- The experiment has been criticized for its lack of informed consent and the emotional distress caused to Albert. Today, such experiments would violate ethical guidelines.
- There is no evidence that Albert's conditioned fear was ever extinguished, potentially leaving him with long-term emotional effects.

Classical Conditioning (Respondent Conditioning) by Ivan Pavlov

Classical conditioning, also known as respondent conditioning, is a learning process discovered by Russian physiologist **Ivan Pavlov** in the early 20th century. It is a fundamental theory in psychology that explains how a neutral stimulus can elicit a conditioned response after being paired with an unconditioned stimulus repeatedly. This learning process forms the basis for many behaviors, both in humans and animals. Classical conditioning involves the formation of an association between two stimuli. A previously neutral stimulus, when paired with a stimulus that naturally elicits a response, comes to trigger a similar response on its own. This process is involuntary and focuses on automatic, reflexive behaviors rather than conscious actions.

Pavlov's Experiment on Classical Conditioning – Pavlov's dogs Experiment

Ivan Pavlov initially conducted research on digestion in dogs, specifically studying their salivation response when food was presented. During these studies, he noticed that dogs would begin to salivate even before food was given—when they heard the footsteps of the lab assistant or saw the food dish. Intrigued by this phenomenon, Pavlov designed an experiment to systematically investigate the association between stimuli and responses.

The Experiment

Pavlov's experiment can be broken down into several stages:

1. **Before Conditioning**

   - Pavlov presented food to the dogs.
   - The food naturally caused the dogs to salivate.

- Pavlov rang a bell. The sound of the bell did not elicit any salivation response from the dogs initially.

2. **During Conditioning**
    - Pavlov repeatedly paired the sound of the bell with the presentation of food
    - Over time, the dogs began to associate the bell with the arrival of food.
3. **After Conditioning**
    - The bell, which was initially a neutral stimulus, became a **Conditioned Stimulus (CS)** after repeated pairings with food.
    - The dogs now salivated (Conditioned Response, CR) when they heard the bell, even if no food was presented (see figure 2).

Findings

- Pavlov demonstrated that behaviors can be conditioned through the association of stimuli.
- The experiment revealed that the learning process involves creating links between previously unrelated stimuli.

Key Terms in Classical Conditioning

1. **Unconditioned Stimulus (UCS):** A stimulus that naturally and automatically triggers a response without prior learning (e.g., food).
2. **Unconditioned Response (UCR):** The natural, automatic response to the UCS (e.g., salivation in response to food).
3. **Neutral Stimulus (NS):** A stimulus that initially does not elicit any response on its own (e.g., the sound of a bell before conditioning).
4. **Conditioned Stimulus (CS):** A previously neutral stimulus that, after being paired with the UCS, elicits a response (e.g., the bell after conditioning).
5. **Conditioned Response (CR):** The learned response to the CS, similar to the UCR (e.g., salivation in response to the bell).

Principles of Classical Conditioning

1. **Acquisition:** The initial stage where the neutral stimulus is paired with the unconditioned stimulus, leading to the conditioned response.
2. **Extinction:** If the conditioned stimulus is repeatedly presented without the unconditioned stimulus, the conditioned response diminishes over time.
3. **Spontaneous Recovery:** After a period of extinction, the conditioned response may reappear when the conditioned stimulus is presented again.

4. **Generalization:** A stimulus similar to the conditioned stimulus may also elicit the conditioned response. For example, a dog conditioned to respond to a bell might also respond to a buzzer.
5. **Discrimination:** The ability to differentiate between the conditioned stimulus and other stimuli that do not predict the unconditioned stimulus.

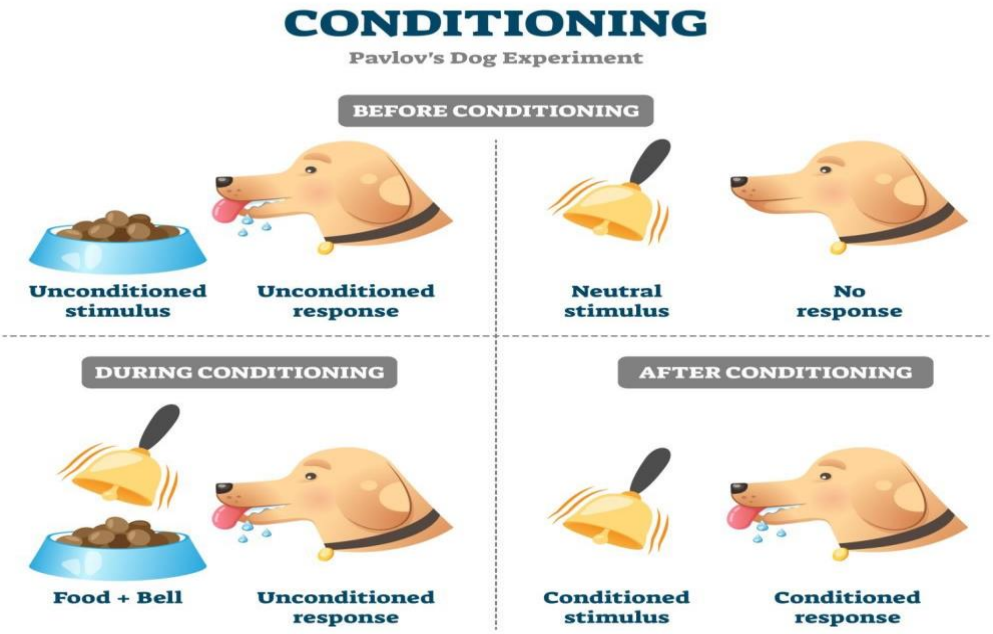


Figure 2: Pavlov's Dog Experiment

Classroom Implication for ICT teachers

1. Teachers can pair ICT lessons with positive stimuli, such as engaging multimedia, interactive activities, or rewards, to foster a favorable attitude toward learning. For instance, using visually appealing software or gamified learning tools can make students associate ICT classes with fun and excitement.
2. Negative behaviors, such as unauthorized browsing or gaming during lessons, can be reduced by pairing these behaviors with mild corrective actions (e.g., losing privileges to use ICT resources). Over time, students learn to associate such misuse with undesirable outcomes.
3. Pairing complex ICT concepts with familiar, repeated stimuli (such as relatable real-world examples) helps reinforce understanding. For example, associating commands in a programming language with everyday actions (like "if-then" logic) makes the material more relatable and easier to recall.